Amish Singh

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EDUCATION

University of Warwick **Coventry, UK**
Master of Mathematics (MMath) 23rd Sep 2024 – 1st Jul 2028

- Grades: On track for First Class Honours – 87.4% (First Year), 82.6% (Second Year)
- Highlighted Modules – Analysis II (93%), Algebra II (92%), Numerical Methods and Computing (92%).

St. Olave's Grammar School **London, UK**
Secondary Education & Sixth Form Sep 2017 – Jul 2024

- A-Levels: Mathematics (A*), Further Mathematics (A*), Physics (A*), Religious Studies (A)
- AS-Levels : Computer Science (a)
- Activities: Sixth Form Association Treasurer, Volleyball Team Captain, Gold Kangaroo SMC

TECHNICAL EXPERIENCE

Real-Option Valuation and Trading Signal Framework for Grey-Green Hydrogen Spread **London, UK**
Independent Research | www.github.com/amsih/Hydrogen-Electrolyser-Project Jun 2026 - Present

- Built a Longstaff-Schwartz least-squares Monte Carlo engine pricing a switching real option on grey/green hydrogen production; validated to exact convergence against a greedy-policy benchmark, with Monte Carlo SE tightening from ±720 to ±295 across 5K→30K simulated paths.
- Calibrated a correlated jump-diffusion model (gas, power, EUA) on 500K+ observations across 7 years of data; regression (N=62,064, R^2=0.86) showed gas-power pass-through more than doubling under stress, with state-dependent correlation rising from 0.20 to 0.26 and jump co-occurrence 7-8x above independence.
- Decomposing plant value via a fixed-commitment, single-switch (Margrabe), full-flexibility and perfect-foresight bridge to isolate the €-value of operational switching flexibility.
- Designing a trading signal from the model's flexibility premium (full-model vs. Margrabe value) rather than raw spread divergence; back testing across 3 energy-shock episodes incl. 2026 Iran-Hormuz crisis.

QuantChallenge 2025 **London, UK**
Research and Trading Competition | 59/1425 (Top 4%) Research Globally Sep 2025

- Invited to a highly selective 3-day Quantitative Research and Trading competition
- Used Principal Component Analysis, Random Forest Regression, and Linear Regression with K-means outlier removal to analyse obfuscated market databases of 16 variables and 80,000 entries to predict 2 targets.
- Obtained an R^2 score of 0.6995, and placed 59th in research out of 1425 teams with an early peak at rank 14.

OTHER EXPERIENCE

Deutsche Bank Investment Banking Work Experience Week **London, UK**
Student Ambassador Jul 2026

- Selected for a competitive 5-day program exploring investment banking in Deutsche Bank.
- Built relationships with 10+ professionals across the investment banking departments, and explored credit and equity sales and trading through informal 1:1 meetings.

Warwick Pride Society **Coventry, UK**
Vice President Feb 2025 – Feb 2026

- Represented society of 400+ members to Warwick Students' Union, alongside the society President(s).
- Organised and produced agendas for 10+ weekly meetings with over 15 regular attendants.

SKILLS & INTERESTS

Technical Skills:	Python (NumPy, Pandas, Matplotlib, SciPy, SKLearn, statsmodels), LaTeX, Stochastic Processes, Monte-Carlo Methods
Certifications:	Kaggle Machine Learning, Kaggle Time Series, JPMorgan Quant Research Job Simulation